

Smart Attendance & Audit Management System

Assignment Track: Assignment 1 - Smart Attendance Management | **Company:** Edumerge Solutions
Submission Deadline: 9:00 AM, 25 September 2026 | **Author:** Harshitha

1. Executive Summary & Problem Framing

Institutional attendance management across 5,000 students and 200 faculty members is a compliance and retention challenge. This solution mitigates 4 core bottlenecks:

- **Instructional Overhead:** Rapid slot-based recording in under 90 seconds.
- **Data Tampering:** Immutable transaction logs replace editable spreadsheets.
- **Dispute Bottlenecks:** Formal dispute desk (PENDING -> APPROVED/REJECTED) with audit trail.
- **Early Defaulter Interventions:** Continuous aggregation identifying students below 75%.

2. Relational Schema & Integrity Constraints

Entity / Table	Key Schema Definition	Integrity Rule / Constraint
sessions	id, subject_id, section_id, faculty_id, date, slot	UNIQUE(subject_id, section_id, date, slot) prevents duplicate lecture logs.
records	id, session_id, student_id, status	UNIQUE(session_id, student_id) prevents double marking.
tickets	id, record_id, student_id, reason, status, reviewed_by	State machine: PENDING -> APPROVED REJECTED.
audit_logs	id, record_id, prev_status, new_status, changed_by, ts	Append-only audit trail capturing all approved mutations.

3. Edge Cases & Engineering Decisions

- **Double Submission Prevention:** Database-level unique composite keys reject duplicate calls with HTTP 400.
- **Division-by-Zero Handling:** Aggregation query handles new cohorts via *HAVING total_classes > 0*.
- **Dispute Cut-off:** Enforces a strict 5-day correction window post-lecture.

4. Mandatory AI Usage Report

AI Tool Used	Gemini / Cursor
Key Prompt	"Design a relational database schema and API endpoints for a college attendance system with 5,000 students that handles attendance co
AI Generated Code	Baseline SQLite schema, FastAPI JWT routes, Tailwind CSS dashboard skeleton.
Candidate Modifications	Wrapped updates in strict atomic database transactions; enforced role-based checks for dispute resolution.
AI Error Rectification	AI wrote INNER JOIN query excluding students with 0 classes from defaulter lists. Identified via unit testing and corrected using LEFT JO